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Multigranularity Information Fusion for Multimodal Retrieval in Prefabricated Construction

Yinyi Wei, Xiao Li, *Member, IEEE*, Yanchen Liu, Ziyang Lin, Wentao Zhu, and Wei Pan

***Abstract*—Prefabricated construction documents are characterized by domain-specific terminology, multilingual content, and rich multimodal elements including texts, tables, and images. General-purpose embedding models exhibit degraded performance in this vertical domain, and existing retrieval approaches often lack explicit cross-granularity alignment between page- and element-level objectives, failing to exploit their complementary strengths. To address these limitations in industrial informatics, this article proposes a multigranularity fusion (MGF) framework for representation learning. A multimodal and multilingual prefabricated construction knowledge base (PCKB) is first constructed from real-world prefabricated construction documents, featuring query-answer pairs explicitly linked to both page- and element-level sources. The MGF framework jointly optimizes three contrastive objectives: a page-level loss to preserve contextual coherence, an element-level loss to enhance fine-grained semantic fidelity, and an answer-guided auxiliary loss to align representations with downstream relevance. Through extensive experiments with multiple embedding models, consistent improvements in retrieval performance are observed after domain-specific training. Notably, the explicit fusion and alignment of page- and element-level objectives yield improvements, while the answer-guided loss enhances element-based retrieval and generally boosts overall performance. Finally, a retrieval module is implemented in a practical system that supports interactive switching between page- and element-level results for both PCKB and user-uploaded documents.**

***Index Terms*—Contrastive learning, domain-specific embedding, industrial informatics, multimodal retrieval, vision-language model, prefabricated construction.**

I. Introduction

PREFABRICATED construction involves manufacturing building components in a controlled factory environment before on-site assembly. It is increasingly adopted globally for enhancing efficiency, reducing labor and waste, and improving quality [1]. The market is projected to grow from USD 173.5 billion in 2025 to USD 302.0 billion by 2035, at a 5.7% CAGR [2].

The growth has fueled complex, multimodal, and multilingual documentation involving diverse stakeholders [3]. Construction professionals spend about 35% of their time searching for information [4], underscoring the need for effective retrieval from heterogeneous sources [5]. Traditional keyword searches falter with natural language, visuals, and multilingual content, while general embedding models suffer domain shift in prefabricated construction [6], [7].

Recent advances in large language models (LLMs) (i.e., foundation models trained on vast text corpora) and multimodal large language models (MLLMs) (i.e., LLM extensions that integrate different modalities into a unified architecture) enable more capable multimodal retrieval. Current methods generally support two retrieval granularities [see Fig. 1(a)]: page-based retrieval encodes entire pages [8], while element-based retrieval embeds fine-grained components separately [9]. Although some models may incorporate both data types during training, they lack explicit cross-granularity alignment between page instances and constituent elements, limiting the exploitation of complementary strengths (e.g., page-level contextual coherence and element-level semantic fidelity).

Despite various models developed for general multimodal retrieval, their effectiveness in prefabricated construction remains unverified. To understand their real-world behavior, we evaluated eight representative state-of-the-art multimodal embedding models on prefabricated construction documents. The investigation in Fig. 1(b) reveals three key findings. 1) All models underperformed relative to public benchmarks, confirming significant domain shift, which is consistent with prior studies [6], [7]. 2) Models trained solely on page-level data (i.e., Nomic, UniSE, and Amann) achieved element-based retrieval performance comparable to, and in some cases superior to, models fine-tuned on mixed data types, suggesting potential cross-granularity knowledge transfer. 3) Page-based retrieval outperformed element-based retrieval for images and tables (due to contextual information),

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Code is available online at https://github.com/CI3LAB/MGF-PC.

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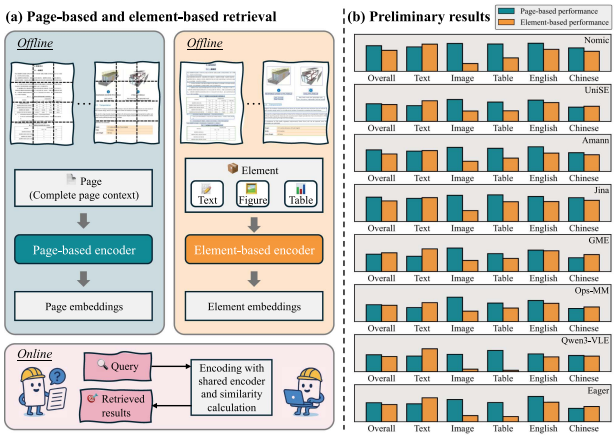


Fig. 1. Introduction of retrieval paradigms, including (a) two retrieval granularities of multimodal retrieval and (b) preliminary NDCG@5 results for multimodal retrieval in prefabricated construction documents.

while element-based retrieval was better for pure text (avoiding visual noise).

These observations indicate that neither granularity is superior; rather, their strengths are modality-dependent and may be complementary. However, this hypothesis remains unexplored due to the lack of: 1) datasets with queries explicitly linked to both pages and elements from the same document, enabling their supervision signals to be jointly aligned and 2) domain-specific studies in fields, such as prefabricated construction, where rich page-element semantics are crucial but underexplored.

To bridge this gap in industrial informatics, where structured knowledge representation and efficient information retrieval are vital for digitalized workflows, this article aims to fuse multigranularity information from both page and element levels for multimodal retrieval in prefabricated construction by encoding pages and elements into dense representations in a shared embedding space. The main contributions are as follows.

1) The first multimodal prefabricated construction knowledge base (PCKB), featuring query–answer pairs generated with human-in-the-loop validation.
2) A multigranularity fusion (MGF) framework, developed to explicitly align page- and element-level information with an answer-guided auxiliary loss, enhanced by lightweight granularity-specific projectors.
3) A retrieval module, deployed practically to support querying the PCKB or user-uploaded documents.

The rest of this article is organized as follows. Section II introduces the background. Section III details the construction of the PCKB. Section IV presents the MGF framework. Section V reports experiments and analyses and Section VI discusses key insights and limitations. Finally, Section VII concludes this article.

II. Related Work

This section reviews three areas that motivate our work. We begin with the evolution of vision-language models, then turn to multimodal retrieval methods with a focus on retrieval granularity, and finally to document analysis in construction engineering. These threads are synthesized in our point of departure, which identifies the key gaps this work addresses.

A. Vision-Language Models

Early vision-language models (VLMs) were driven by contrastive learning objectives (e.g., contrastive language-image pre-training (CLIP) [21]), learning joint image-text representations via large-scale pretraining on image captioning and visual question answering. While effective for coarse-grained tasks, these dual-encoder architectures are constrained by fixed token lengths and struggle with layout-rich or text-dense documents [8]. Subsequent work on MLLMs (e.g., LLaVA [22] and Qwen-VL series [23]) integrated powerful LLM decoders with visual encoders, enabling richer cross-modal reasoning and longer context understanding. Concurrently, document-oriented VLMs, such as LayoutLMv3 [24], addressed structured document understanding but were not designed for end-to-end retrieval. Recent retrieval-focused extensions of MLLMs have demonstrated that pooled MLLM representations substantially outperform CLIP-style embeddings on multimodal retrieval benchmarks [8], [9], [16], [20], motivating our approach.

B. Multimodal Retrieval

Multimodal retrieval seeks to map heterogeneous data in a shared space [25]. Early CLIP-based methods fail on technical documents due to short context windows and loss of visual details. Thus, research has shifted toward MLLM-based retrieval, with two retrieval granularities: page- and element-based. Page-based retrieval treats a whole page as a retrieval unit, preserving layout and context but limiting fine-grained localization and potentially including irrelevant content [8]. Element-based retrieval decomposes pages into fine-grained units, improving semantic precision but risking context fragmentation [9]. As summarized in Table I, earlier element-based methods typically represent content through image-caption pairs, a strategy constrained by short token limits and outdated architectures. The advent of page-based retrieval introduced an alternative by encoding entire pages holistically, influencing subsequent retrieval model designs. However, existing models have yet to explicitly fuse page- and element-level signals, leaving complementary cross-granularity signals underutilized. Moreover, existing models are mostly trained on general-purpose data, limiting their performance in specialized domains, such as prefabricated construction.

C. Document Analysis in Construction Engineering

The construction sector is highly document-intensive, where over 80% of its data are unstructured or semistructured, including standards, drawings, and regulations [3]. Effective document analysis is crucial for productivity and safety, and decision-making [26], [27]. While early approaches used rule-based or shallow machine learning methods, recent work leverages LLMs for end-to-end, domain-specific understanding [28]. Prefabricated construction intensifies this need by

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TABLE I
REPRESENTATIVE STUDIES ON MULTIMODAL RETRIEVAL APPROACHES

Ref	Year	Method	Retrieval granularity	Key features	Data source
[10]	2023	UniVL-DR	Element-based	Vision-language dual encoder with image verbalization	General web pages
[11]	2024	MARVEL	Element-based	Visual module plugin on any text retriever	General web pages
[12]	2024	VISTA	Element-based	Text encoder with visual tokens; data generation	General web pages and fashion products
[13]	2024	UniIR	Element-based	Supports diverse tasks through multimodal query instruction handling	General web pages, news, and fashion products
[14]	2024	DSE	Page-based	Encoding document screenshots into dense embeddings	General web pages
[8]	2025	ColPali	Page-based	Benchmark construction and direct embedding of document page images	General web pages, reports, and publications
[15]	2025	VisRAG	Page-based	Page images input; comparison across diverse settings	General textbooks, academic papers, and product manuals
[16]	2025	VLM2Vec	Element-based	Universal multimodal embeddings for downstream tasks	General data supporting downstream tasks
[17]	2025	Docopilot	Page-based	Document-level dependencies handling with a new benchmark	Academic papers and general web pages
[18]	2025	VDocRAG	Page-based	Understanding varied documents and modalities in a unified image format	General documents, slides, and web pages
[9]	2025	GME	Both	Balanced multimodal datasets synthesis and construction	General web pages
[19]	2025	UniME	Element-based	Knowledge distillation for diverse downstream tasks	General data supporting downstream tasks
[20]	2025	Jina-embeddings-v4	Both	Universal multimodal multilingual embedding model	General web pages
-	-	This work	Both	Plug-and-play multimodal embedding framework	Prefabricated construction documents

integrating manufacturing precision with on-site coordination, featuring multilingual, multimodal, and highly technical documents across disciplines [1]. However, research on effective retrieval for such content remains scarce.

D. Point of Departure

Despite rapid progress, the following three gaps persist in applying multimodal retrieval to prefabricated construction. 1) General-purpose embedding models underperform on domain-specific, multilingual, and multimodal content.

2) Existing datasets lack coverage of technical and visual information specific to prefabricated construction.

3) The lack of query–answer pairs explicitly grounded in both page- and element-level sources limits the ability of current models to align representations across granularities during optimization, overlooking the complementary signals between these levels.

Motivated by these gaps, the following objectives are set forth: 1) to construct a multimodal and multilingual knowledge base for prefabricated construction (PCKB), containing aligned pages and elements with query-answer pairs; 2) to design a MGF framework that integrates complementary objectives; and 3) to implement a retrieval module in a practical system allowing users to query or upload documents, with flexible switching between page- and element-level results.

III. CONSTRUCTION OF PCKB

To support domain-specific representation learning in prefabricated construction, we built a high-quality, multimodal, and multilingual PCKB. Previous general-purpose retrieval datasets, often based on web-crawled question-passage pairs (e.g., from Wikipedia) [10], [11], [13], suffer from high noise, low domain specificity, context dependency, and unreliable ground truth. Recent advances in MLLMs enable controlled synthetic data generation, producing contextually grounded retrieval data for diverse domains (e.g., general [20], [29], academic [15], [17], and medicine [7]) and data formats (e.g., web pages [9], textbooks [12], slides [18], and reports [8]). These methods prompt MLLMs to generate context-independent questions from passages or visual elements, with outputs further validated by automated models or human reviewers, resulting in cleaner, more diverse, and better-aligned training signals, especially critical in specialized fields with scarce real queries. Accordingly, we adopted this synthetic data generation strategy to construct PCKB from curated real-world documents, covering regionally diverse prefabrication practices, such as panelized and modular construction systems.

A. Document Collection and Preprocessing

We manually collected real-world prefabricated construction documents (including technical specifications, construction manuals, regulatory codes, reports, and guidelines) in PDF format from authoritative sources across diverse geographical regions to ensure domain coverage and multilingual representation. For each document, we removed its table of contents and references and manually filtered meaningless images and tables. The raw corpus $\mathcal{D}_{\text{raw}}$ consists of

$$\mathcal{D}_{\text{raw}} = \{d_1, d_2, \cdots, d_{|\mathcal{D}_{\text{raw}}|}\}, \quad |\mathcal{D}_{\text{raw}}| = 95 \tag{1}$$

where each document d_i is sourced from one of five regions: Mainland China (52), Hong Kong (26), North America (5), Europe (7), and Singapore (5). The collection comprises 43 English documents and 52 Chinese documents.

Each document d_i is decomposed into individual pages: $d_i = \{p_i^1, p_i^2, \cdots, p_i^{n_i}\}$, where p_i^j represents the jth page of document d_i, and n_i denotes the total number of pages in that document. Across the entire corpus, we obtain

$$\mathcal{P} = \bigcup_{i=1}^{|\mathcal{D}_{\text{raw}}|} \{p_i^1, p_i^2, \ldots, p_i^{n_i}\}, \quad |\mathcal{P}| = 6{,}177 \tag{2}$$

comprising 4018 English pages and 2159 Chinese pages.

To support explicit cross-granularity alignment during training, for each page p, we employed MinerU,[1] an open-source document parsing tool, to extract three types of elements (i.e., texts, images, and tables). Let $\mathcal{E}(p)$ denote the set of elements

$$\mathcal{E}(p) = \{e_p^{\text{text}}\} \cup \{e_p^{\text{img},i}\}_{i=0}^{m_p} \cup \{e_p^{\text{tab},j}\}_{j=0}^{k_p} \tag{3}$$

where e_p^{text} is the textual content of page p (exactly one per page), $\{e_p^{\text{img},i}\}_{i=0}^{m_p}$ are the m_p images extracted from page p, and $\{e_p^{\text{tab},j}\}_{j=0}^{k_p}$ are the k_p tables extracted from page p.

Tables are commonly represented as images or plain text (e.g., HTML or linearized text). Preliminary experiments in Fig. 2 show comparable performance in table-only retrieval for both formats. However, tables in PDF-based or OCR-processed construction documents often include complex features, such as merged cells, color coding, and graphical symbols, which are

[1] [Online]. Available: https://mineru.net/

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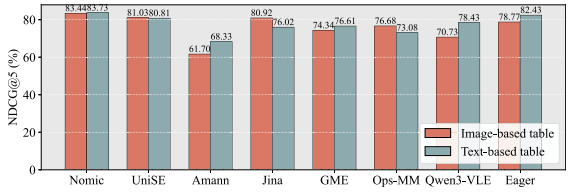


Fig. 2. Comparison of image- and text-based table representations under table-only retrieval in prefabricated construction documents.

challenging to capture with plain text. In addition, table-to-text conversion is nontrivial and unreliable across document types. Therefore, to enhance compatibility and generalizability, we adopted image-based table representations.

The complete element set across all pages is

$$\mathcal{E} = \bigcup_{p \in \mathcal{P}} \mathcal{E}(p) \quad |\mathcal{E}| = 7{,}969 \tag{4}$$

with 5481 text elements, 1065 images, and 1423 tables.

B. Synthetic Query–Answer Pair Generation

Following the synthetic data generation paradigm in retrieval model training, we used an MLLM[2] to produce contextually grounded query–answer pairs for extracted elements.

For each element $e \in \mathcal{E}(p)$ in a page $p \in \mathcal{P}$, we prompted the MLLM to generate a variable number of pairs

$$\text{MLLM}(e, p) \rightarrow \left\{(q_1^e, a_1^e), \cdots, (q_{n_e}^e, a_{n_e}^e)\right\} \tag{5}$$

where $n_e \in \{0, 1, 2, 3\}$ represents the number of query–answer pairs generated for element e, depending on the content's information specificity and richness specified in the prompt.

Prompt design was carefully controlled to ensure: 1) each query is self-contained and context-independent, avoiding vagueness or broadness (e.g., "What is shown in the figure?" or "What does this component connect to?"); 2) each query is answerable from the given element, with fewer generated if information is insufficient; and 3) key descriptors from the document title are incorporated where appropriate to enhance real-world relevance, which is an important consideration when various regions address similar issues.

To ensure quality and alignment, we adopted a two-stage verification process. In addition to employing a separate model to assess context-independence and relevance [7], [9], [15], we conducted manual review and editing. Context-dependent, irrelevant, overly generic, and hallucinated pairs were revised to be context-independent, specific, and factual, or excluded if unrectifiable. Consequently, 1561 image-based pairs were excluded with 866 revised; 433 table-based pairs were excluded with 250 revised; and 529 text-based pairs were excluded with 169 revised. The higher exclusion of image-based pairs underscores the difficulty of extracting context-independent, factual information from less semantically explicit visual content compared to structured tables or text.

[2]We used the application programming interface (API) of GPT-4.1 in our implementation.

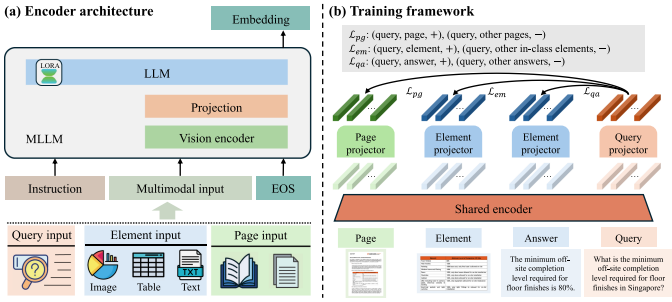


Fig. 3. Overview of MGF: (a) the encoder architecture and (b) the training framework.

The complete knowledge base is formalized as follows:

$$\text{PCKB} = \{(q, a, e, p) \mid e \in \mathcal{E}(p), p \in \mathcal{P}\}. \tag{6}$$

The final PCKB consists of 22 854 query-answer pairs: 16 183 text-based, 2594 image-based, and 4077 table-based.

IV. MGF Framework

Fig. 3 presents the MGF framework. MGF learns a unified embedding space to represent both pages and elements, enabling seamless retrieval across varying levels of granularity without sacrificing performance at either end.

A. Encoder Architecture

As illustrated in Fig. 3(a), the MLLM encoder handles heterogeneous inputs through a unified framework: 1) text-based queries; 2) fine-grained elements including images, tables, and texts; and 3) document pages rendered as images.

The encoding architecture consists of three main modules. The vision encoder, based on a pretrained vision transformer, segments visual inputs into patches, embeds them with positional encodings, and extracts features via transformer layers. A projection layer then maps these visual features into the language model's token space. Finally, the core LLM processes both projected visual tokens and original text tokens using self-attention, with the final hidden state of the last token representing the input for downstream tasks.

B. Multigranularity Training Framework

Fig. 3(b) presents the training framework, in which the encoder and type-specific projectors map the inputs into dense embeddings. During training, multigranularity contrastive objectives are employed to optimize these representations, thereby improving their discriminative power.

1) Encoder and Projectors: To enhance the distinction between different representation types while maintaining a unified embedding space, granularity-specific projectors are introduced. Formally, let f_θ denote the MLLM encoder parameterized by θ, and g_{ϕ_p}, g_{ϕ_e}, and g_{ϕ_q} represent the page projector, element projector, and query projector, respectively. Each projector is implemented as a lightweight, fully connected layer, defined by

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$g_\phi(z) = Wz + b$, where $W \in \mathbb{R}^{d\times d}$ and $b \in \mathbb{R}^d$ are learnable parameters specific to each projector.

Given a training instance (q, a, e, p), projected representations are obtained as follows:

$$z_q = g_{\phi_q}(f_\theta(q)) \qquad z_a = g_{\phi_a}(f_\theta(a)) \tag{7}$$

$$z_e = g_{\phi_e}(f_\theta(e)) \qquad z_p = g_{\phi_p}(f_\theta(p)). \tag{8}$$

2) Multigranularity Contrastive Objectives: We employ three contrastive losses to fine-tune the encoder and projectors.

The page-level loss, $\mathcal{L}_{pg}$, aligns query representations with their corresponding page representations.

For a training batch with N instances $(q_i, a_i, e_i, p_i)_{i=1}^N$, the page-level contrastive loss is

$$\mathcal{L}_{pg} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_{q_i}, z_{p_i})/\tau)}{\sum_{j=1}^{N}\exp\left(\text{sim}\left(z_{q_i}, z_{p_j}\right)/\tau\right)} \tag{9}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity: $\text{sim}(z_1, z_2) = \frac{z_1^T z_2}{\|z_1\|\|z_2\|}$, and τ is a temperature hyperparameter, controlling the sharpness of the distribution. For each query q_i, the page p_i containing its corresponding element is the positive sample, while all other pages $\{p_j\}_{j\neq i}$ are in-batch negative samples.

The element-level loss, $\mathcal{L}_{em}$ aligns queries with their specific source elements

$$\mathcal{L}_{em} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_{q_i}, z_{e_i})/\tau)}{\sum_{j=1}^{N}\exp\left(\text{sim}\left(z_{q_i}, z_{e_j}\right)/\tau\right)} \tag{10}$$

where each query q_i is paired with its originating element e_i as the positive sample, while other in-class elements $\{e_j\}_{j\neq i}$ (i.e., elements of the same type) serve as in-batch negatives.

To deepen domain familiarity, we introduce an answer-guided contrastive loss $\mathcal{L}_{qa}$, aligning queries with their corresponding ground-truth answers

$$\mathcal{L}_{qa} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_{q_i}, z_{a_i})/\tau)}{\sum_{j=1}^{N}\exp\left(\text{sim}\left(z_{q_i}, z_{a_j}\right)/\tau\right)} \tag{11}$$

where each query q_i and its associated answer a_i form the positive pair, while answers to other queries serve as negatives.

The overall training objective combines the three losses

$$\mathcal{L}_{\text{total}} = \lambda_{pg}\mathcal{L}_{pg} + \lambda_{em}\mathcal{L}_{em} + \lambda_{qa}\mathcal{L}_{qa} \tag{12}$$

where λ_{pg}, λ_{em}, and λ_{qa} are hyperparameters controlling the relative importance. All losses flow back through the shared encoder and projectors for multimodal representation learning.

3) Increasing Batch Size via GradCache: Collecting sufficient hard negatives is difficult or ambiguous for multimodal datasets. Increasing batch size helps by providing more in-batch negatives for contrastive learning, thus improving training efficiency. However, graphics processing unit (GPU) memory limits batch size, especially in the construction industry. PCKB's complex inputs (e.g., images and text per instance) exacerbate memory consumption. Gradient accumulation, commonly used to simulate large batches, is incompatible with contrastive learning, where each instance must interact with all others in the batch.

TABLE II
LANGUAGE DISTRIBUTION OF THE EVALUATION DATASET

Split	File			Page			Element		
	EN	CN	All	EN	CN	All	EN	CN	All
Train	31	35	66	3,458	1,544	5,002	4,132	2,164	6,296
Test	12	17	29	560	615	1,175	797	876	1,673
All	43	52	95	4,018	2,159	6,177	4,929	3,040	7,969

TABLE III
DISTRIBUTION OF ELEMENT TYPES IN THE EVALUATION DATASET

Split	Element type count				Query type count			
	Image	Table	Text	All	Image	Table	Text	All
Train	681	1,134	4,481	6,296	1,658	3,291	13,245	18,194
Test	384	289	1,000	1,673	936	786	2,938	4,660
All	1,065	1,423	5,481	7,969	2,594	4,077	16,183	22,854

TABLE IV
DISTRIBUTION OF QUERY-ANSWER PAIR LENGTHS IN THE EVALUATION DATASET, REPORTED AS AVERAGE/MINIMUM/MAXIMUM TOKEN COUNTS

Split	Query			Answer		
	Image	Table	Text	Image	Table	Text
Train	20.9/9/48	23.6/9/62	19.8/5/47	33.7/5/129	30.8/1/206	61.5/2/496
Test	20.9/9/54	22.0/9/52	19.3/5/45	32.0/6/129	31.6/1/158	52.9/7/241
All	20.9/9/54	23.3/9/62	19.7/5/47	33.1/5/129	31.0/1/206	59.9/2/496

To address this limitation, we adopted GradCache [30], which separates the computation of contrastive loss gradients from the encoder's backward pass, removing the need to store full-batch activations. Specifically, a large batch $\mathcal{B}$ is split into smaller sub-batches $\mathcal{B} = \{\widehat{\mathcal{B}}_1, \widehat{\mathcal{B}}_2, \ldots, \widehat{\mathcal{B}}_S\}$ that individually fit on a single GPU. First, for every subbatch, the per-sample gradient tensors are computed and stored: $u_i = \frac{\partial\mathcal{L}}{\partial z_i}$. Then, these cached gradient tensors are reused to compute the gradients of the parameters of the encoder and projectors by accumulating contributions across all subbatches

$$\frac{\partial\mathcal{L}}{\partial\Theta} = \sum_{\widehat{B}_j\in\mathcal{B}}\sum_{z_i}\frac{\partial\mathcal{L}}{\partial z_i}\frac{\partial z_i}{\partial\Theta} = \sum_{\widehat{B}_j\in\mathcal{B}}\sum_{z_i}u_i\frac{\partial z_i}{\partial\Theta} \tag{13}$$

where z_i denotes an instance embedding in the subbatch $\widehat{B}_j$, and Θ represents all trainable parameters.

V. EXPERIMENTAL EVALUATIONS AND SYSTEM DEMONSTRATIONS

A. Experimental Setup

Following established protocols and to facilitate comparative evaluation [8], [9], we partitioned the documents in PCKB into training and test sets at an 8:2 ratio. The distributions of language, element types, and query-answer pair lengths are summarized in Tables II–IV, respectively.

Since MGF is designed to be plug-and-play on multimodal embedding models rather than a standalone retrieval system, we evaluate it by substituting the backbone across a set of competitive, publicly validated models. Specifically, we implemented

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and evaluated it atop eight state-of-the-art models: 1) Nomic;[3] 2) UniSE;[4] 3) Amann;[5] 4) Jina;[6] 5) GME;[7] 6) Ops-MM;[8] 7) Qwen3-VLE;[9] and 8) Eager.[10] The first three were trained solely on page-level data, whereas the remaining five models were primarily trained on element-level data, supplemented with a minor proportion of page-level data. Model selection followed two criteria: 1) the backbone size should not exceed four billion parameters to facilitate resource-efficient fine-tuning and lightweight deployment in practical applications; and 2) the model must demonstrate basic competency in understanding prefabricated construction data. Through extensive pilot assessments, we excluded models that consistently performed poorly in this specialized domain. For example, a CLIP-based encoder developed by BAAI General Embeddings (BGE-VL-CLIP) was eliminated due to its inability to process long sequences, and VLM2Vec was discarded owing to subpar retrieval performance (below 1% on NDCG@5) attributed to excessive downstream task initialization.

A validation set comprising 5% of the training samples was set aside for hyperparameter selection, following the common practice in prior methods [8], [9]. Model training used a learning rate of 3e-4 for encoders and 1e-4 for projectors, with a temperature of 0.02. GradCache was utilized to achieve an effective batch size of 128, with subbatch sizes configured as follows: two for Jina, and four for the remaining models. The weights for the three losses were set to 1:1:1. During experimentation, we observed that simultaneous training of both the encoder and the projectors often led to instability.[11] To mitigate this, we trained the encoder for one epoch, followed by the projectors for another while keeping the encoder parameters fixed. Projectors were initialized as identity mappings to preserve the integrity of the embedding space. For image-based inputs, images were resized so that the longer side did not exceed 768 pixels, and the maximum number of image tokens was set to 384, as discussed in Section V-D. MGF was implemented atop these general-purpose embedding models to leverage their broad retrieval capabilities, which stem from large-scale pretraining on general-domain data. All models were fine-tuned using bfloat16 precision and low-rank adapters with alpha and rank value set to 32. The average results across three runs were reported. Training was performed on a 2-GPU setup (NVIDIA GeForce RTX 4090, 48 GB each), utilizing the accelerate library for efficient data parallelism. Fine-tuning and evaluation required approximately 8 h for a 4-B model and 4 h for a 2-B model.

To evaluate the retrieval performance, we employed standard metrics following prior research [8], [10]: normalized discounted cumulative gain (NDCG@K), mean reciprocal rank (MRR@K), and recall@K. NDCG@K is a position-sensitive metric that assigns higher scores to relevant items appearing earlier in the ranking list. MRR@K is calculated as the average of the reciprocal ranks of the identified item across all queries. Recall@K measures the fraction of queries for which the ground-truth item appears within the top-K retrieved results.

TABLE V
PAGE-BASED RETRIEVAL RESULTS UNDER FOUR SETTINGS: 1) ORIGINAL (NO FINE-TUNING); 2) PAGE-LEVEL FINE-TUNING; 3) ELEMENT-LEVEL FINE-TUNING; AND 4) MGF-BASED FINE-TUNING.

Model	Method	NDCG@5 (%)	MRR@5 (%)	Recall@5 (%)
Nomic	Original	69.20	64.80	82.34
	Fine-tuned (Page)	<u>78.83</u>	<u>74.68</u>	<u>91.14</u>
	Fine-tuned (Element)	77.10	72.96	89.40
	MGF	**80.42**	**76.27**	**92.73**
UniSE	Original	58.21	53.28	73.03
	Fine-tuned (Page)	<u>77.47</u>	<u>73.13</u>	<u>90.34</u>
	Fine-tuned (Element)	73.51	69.26	86.16
	MGF	**79.30**	**75.14**	**91.63**
Amann	Original	47.57	43.12	60.97
	Fine-tuned (Page)	<u>74.03</u>	<u>69.46</u>	<u>87.64</u>
	Fine-tuned (Element)	71.42	66.89	84.94
	MGF	**76.90**	**72.52**	**89.89**
Jina	Original	65.81	61.25	79.46
	Fine-tuned (Page)	<u>77.68</u>	<u>73.50</u>	<u>90.06</u>
	Fine-tuned (Element)	74.61	70.36	87.25
	MGF	**78.78**	**74.47**	**91.55**
GME	Original	47.03	42.89	59.53
	Fine-tuned (Page)	<u>74.22</u>	<u>69.73</u>	<u>87.60</u>
	Fine-tuned (Element)	68.19	63.77	81.39
	MGF	**74.82**	**70.24**	**88.43**
Ops-MM	Original	45.65	41.22	59.01
	Fine-tuned (Page)	<u>74.46</u>	<u>69.94</u>	<u>87.90</u>
	Fine-tuned (Element)	66.72	62.02	80.75
	MGF	**74.64**	**70.12**	**88.07**
Qwen3-VLE	Original	45.68	41.33	58.80
	Fine-tuned (Page)	<u>77.82</u>	<u>73.52</u>	<u>90.60</u>
	Fine-tuned (Element)	75.23	70.99	87.81
	MGF	**79.14**	**74.87**	**91.76**
Eager	Original	51.33	47.24	63.58
	Fine-tuned (Page)	<u>75.52</u>	<u>71.19</u>	<u>88.43</u>
	Fine-tuned (Element)	71.64	67.40	84.29
	MGF	**77.71**	**73.42**	**90.49**

For each model, bold and underline indicate the best and second-best values, respectively.

B. Main Results

Tables V and VI report the performance of various models on page- and element-based retrieval tasks under four conditions: original pretrained, fine-tuned with the page-level objective, fine-tuned with the element-level objective, and fine-tuned within MGF. The results yield three key observations given as follows. 1) All models achieved better results after fine-tuning on either page- or element-level data, underscoring the insufficiency of general-purpose embeddings for prefabricated construction. 2) MGF-tuned models outperformed those using standard fine-tuning methods (e.g., performance gains ranging from 0.3% to 4.4% in page-based retrieval following page-level training, and 1.3% to 5.2% in element-based retrieval following element-level training), validating MGF's effectiveness in fusing multigranularity information. 3) MGF provided mutually

[3][Online]. Available: https://huggingface.co/nomic-ai/nomic-embed-multimodal-3b
[4][Online]. Available: https://huggingface.co/BAAI/BGE-VL-Screenshot
[5][Online]. Available: https://huggingface.co/racineai/QwenAmann-4B-dse
[6][Online]. Available: https://huggingface.co/jinaai/jina-embeddings-v4
[7][Online]. Available: https://huggingface.co/Alibaba-NLP/gme-Qwen2-VL-2B-Instruct
[8][Online]. Available: https://huggingface.co/OpenSearch-AI/Ops-MM-embedding-v1-2B
[9][Online]. Available: https://huggingface.co/Qwen/Qwen3-VL-Embedding-2B
[10][Online]. Available: https://huggingface.co/eagerworks/eager-embed-v1
[11]This behavior can be attributed to an optimization mismatch between the stable, pretrained representations of the large model and the fully connected layer. End-to-end training introduces noisy gradients from the head that can disrupt the model's learned semantic structure. A two-stage training strategy decouples these objectives, improving optimization stability.

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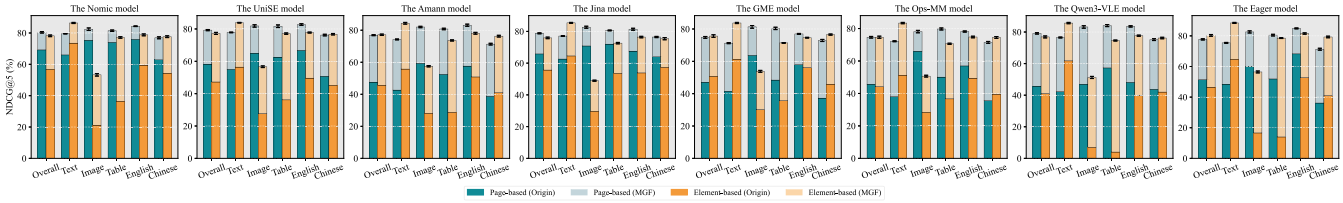


Fig. 4. NDCG@5 (%) of original models and MGF-tuned models across element types and languages under page- and element-based retrieval settings with error bars indicating standard deviation.

TABLE VI
ELEMENT-BASED RETRIEVAL RESULTS UNDER FOUR SETTINGS: 1) ORIGINAL (NO FINE-TUNING); 2) PAGE-LEVEL FINE-TUNING; 3) ELEMENT-LEVEL FINE-TUNING; AND (4) MGF-BASED FINE-TUNING

Model	Method	NDCG@5 (%)	MRR@5 (%)	Recall@5 (%)
Nomic	Original	56.74	52.69	68.84
	Fine-tuned (Page)	71.34	67.25	83.48
	Fine-tuned (Element)	75.50	71.56	87.17
	MGF	**78.18**	**74.58**	**88.84**
UniSE	Original	47.21	42.18	62.40
	Fine-tuned (Page)	72.94	68.77	85.34
	Fine-tuned (Element)	75.27	71.43	86.61
	MGF	**77.25**	**73.40**	**88.63**
Amann	Original	45.60	41.23	58.76
	Fine-tuned (Page)	66.50	62.14	79.48
	Fine-tuned (Element)	75.30	71.27	87.25
	MGF	**77.15**	**73.20**	**88.82**
Jina	Original	55.64	51.37	68.39
	Fine-tuned (Page)	72.31	68.37	83.99
	Fine-tuned (Element)	73.88	69.75	86.12
	MGF	**75.90**	**72.12**	**87.10**
GME	Original	50.69	46.11	64.42
	Fine-tuned (Page)	70.07	65.90	82.47
	Fine-tuned (Element)	72.38	68.19	84.85
	MGF	**75.63**	**71.73**	**87.19**
Ops-MM	Original	44.15	39.53	58.09
	Fine-tuned (Page)	67.98	63.57	81.18
	Fine-tuned (Element)	72.11	67.86	84.74
	MGF	**74.65**	**70.71**	**86.33**
Qwen3-VLE	Original	40.96	37.37	51.70
	Fine-tuned (Page)	68.42	64.31	80.67
	Fine-tuned (Element)	75.94	72.19	87.41
	MGF	**76.89**	**73.13**	**88.00**
Eager	Original	46.45	42.71	57.66
	Fine-tuned (Page)	69.72	65.53	82.17
	Fine-tuned (Element)	79.16	75.63	89.94
	MGF	**80.34**	**76.95**	**90.36**

For each model, bold and underline indicate the best and second-best values, respectively.

reinforcing cross-granularity improvement [e.g., MGF-based fine-tuning obtained improvements of 5.5–17.8% in element-level retrieval (versus page-level training) and 4.5–13.1% in page-level retrieval (versus element-level training)], indicating that MGF not only enhances domain-specific knowledge acquisition but also promotes knowledge transfer between different granularities.

Fig. 4 illustrates the performance improvement across element types and languages. The error bars, indicating the standard deviation, highlight the consistency of enhancements.

In addition, Fig. 5 shows that as the retrieval item K increases from 1 to 100, retrieval performance improved rapidly at first, but began to plateau as K exceeded 10.

C. Ablation Study

To understand the contribution of each component to the overall performance, a series of ablation studies was conducted.

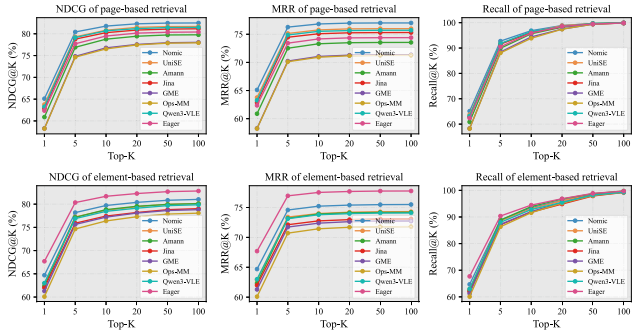


Fig. 5. Performance of different MGF-tuned models across top-K settings in page- and element-based retrieval.

TABLE VII
EFFECT OF LOSSES ON NDCG@5 (%) IN PAGE-BASED RETRIEVAL

Setting	Nomic	UniSE	Amann	Jina	GME	Ops-MM	Qwen3-VLE	Eager
$\mathcal{L}_{pg}+\mathcal{L}_{em}+\mathcal{L}_{qa}$	**80.42**	**79.30**	**76.90**	**78.78**	**74.82**	**74.64**	**79.14**	**77.71**
$\mathcal{L}_{pg}+\mathcal{L}_{em}$	79.92	77.99	74.90	78.60	73.61	74.55	78.35	76.56
$\mathcal{L}_{pg}+\mathcal{L}_{qa}$	79.47	77.17	75.37	78.48	74.12	73.93	77.45	76.66
$\mathcal{L}_{em}+\mathcal{L}_{qa}$	75.21	72.19	72.67	74.64	68.37	64.23	69.95	72.45
$\mathcal{L}_{pg}$	78.83	77.47	74.03	77.68	74.22	74.46	77.82	75.52
$\mathcal{L}_{em}$	77.10	73.51	71.42	74.61	68.19	66.72	75.23	71.64
$\mathcal{L}_{qa}$	66.73	61.00	64.11	65.42	52.48	52.61	61.61	62.48
Origin	69.20	58.21	47.57	65.81	47.03	45.65	45.68	51.33

For each model, bold and underline indicate the best and second-best values, respectively.

TABLE VIII
EFFECT OF LOSSES ON NDCG@5 (%) IN ELEMENT-BASED RETRIEVAL

Setting	Nomic	UniSE	Amann	Jina	GME	Ops-MM	Qwen3-VLE	Eager
$\mathcal{L}_{pg}+\mathcal{L}_{em}+\mathcal{L}_{qa}$	78.18	**77.25**	**77.15**	75.90	75.63	74.65	**76.89**	**80.34**
$\mathcal{L}_{pg}+\mathcal{L}_{em}$	76.48	75.48	73.73	74.35	72.95	73.04	76.05	76.69
$\mathcal{L}_{pg}+\mathcal{L}_{qa}$	74.00	70.98	73.46	72.26	73.30	70.20	69.82	77.93
$\mathcal{L}_{em}+\mathcal{L}_{qa}$	**78.45**	75.37	76.67	**76.37**	**75.83**	**75.19**	76.08	80.11
$\mathcal{L}_{pg}$	71.34	72.94	66.50	72.31	70.07	67.98	68.42	69.72
$\mathcal{L}_{em}$	75.50	75.27	75.30	73.88	72.38	72.11	75.94	79.16
$\mathcal{L}_{qa}$	70.77	65.67	70.29	66.27	62.95	60.27	68.40	71.96
Origin	56.74	47.21	45.60	55.64	50.69	44.15	40.96	46.45

For each model, bold and underline indicate the best and second-best values, respectively.

In particular, the impact of loss terms, loss coefficients, and projector design was examined.

Tables VII and VIII present the effects of loss components. In page-based retrieval, the full loss combination yielded the highest scores across models, with $\mathcal{L}_{pg}$ contributing most significantly, as its consistent optimization objective captured global semantic coherence. The $\mathcal{L}_{em}$ and $\mathcal{L}_{qa}$ losses introduced finer element-level alignment, further enhancing accuracy. In contrast, fine-grained supervision was more critical in element-based retrieval. The $\mathcal{L}_{\text{qa}}$ loss, when used alongside $\mathcal{L}_{\text{em}}$, consistently improved performance, emphasizing the importance of query-specific and domain-semantic alignment. Notably, models trained with both $\mathcal{L}_{\text{em}}$ and $\mathcal{L}_{\text{qa}}$ sometimes surpass those

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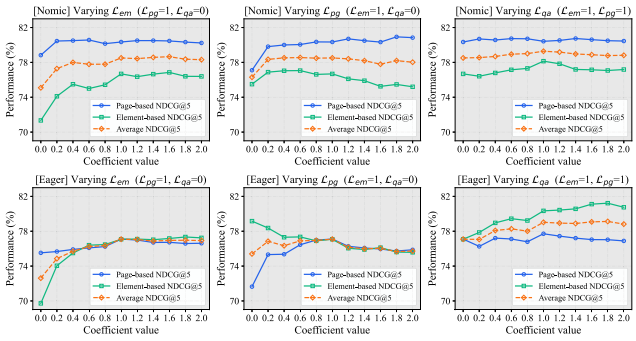


Fig. 6. Effect of loss coefficients on NDCG@5 (%) sensitivity analysis of loss weights in MGF-tuned Nomic and Eager.

TABLE IX
PROJECTOR EFFECTS ON PAGE-BASED NDCG@5 (%)

Setting	Nomic	UniSE	Amann	Jina	GME	Ops-MM	Qwen3-VLE	Eager
MGF	**80.42**	**79.30**	**76.90**	**78.78**	**74.82**	**74.64**	**79.14**	**77.71**
-proj	79.96	78.95	76.54	78.45	74.20	74.06	78.56	77.09
Origin	69.20	58.21	47.57	65.81	47.03	45.65	45.68	51.33

For each model, bold and underline indicate the best and second-best values, respectively.

TABLE X
PROJECTOR EFFECT ON ELEMENT-BASED NDCG@5 (%)

Setting	Nomic	UniSE	Amann	Jina	GME	Ops-MM	Qwen3-VLE	Eager
MGF	**78.18**	**77.25**	**77.15**	**75.90**	**75.63**	**74.65**	**76.89**	**80.34**
-proj	78.14	76.98	76.85	75.65	75.42	74.09	76.33	80.03
Origin	56.74	47.21	45.60	55.64	50.69	44.15	40.96	46.45

For each model, bold and underline indicate the best and second-best values, respectively.

trained with all losses. Nevertheless, the full loss combination remained optimal when considering holistically across both page- and element-based retrieval, which can integrate heterogeneous signals at multiple granularities.

Fig. 6 presents the sensitivity analysis of the Nomic and Eager models with respect to different loss coefficients, from which we observed the following. 1) Incorporating both page- and element-level objectives was essential for achieving more accurate information retrieval. 2) When $\mathcal{L}_{qa}$ was set to zero and $\mathcal{L}_{pg}$ was set to one, the models demonstrated comparable performance when $\mathcal{L}_{em}$ ranged from one to two. Conversely, when $\mathcal{L}_{em}$ was fixed at one, comparable results were observed as $\mathcal{L}_{pg}$ varied between 0.2 and 1. (3) While $\mathcal{L}_{qa}$ contributed less significantly than the other two losses, it still provided additional performance improvements. Based on these observations, we set $\mathcal{L}_{pg}$, $\mathcal{L}_{em}$, and $\mathcal{L}_{qa}$ to one in our experiments to ensure robust and balanced model training.

In addition, as indicated in Tables IX and X, removing projectors from MGF consistently led to a decline in performance across all models and retrieval settings. Although the improvements were slight, the projectors did not add inference latency; therefore, they are retained in our framework.

D. Effect of the Number of Image Tokens

In MLLMs, images are divided into patches, tokenized, and encoded as embeddings for language model processing. Increasing the number of image tokens, enables model to capture finer visual details, enhancing tasks, such as small-object recognition. However, too many tokens can add redundant visual information, potentially distracting models from focusing on key content. In addition, increasing the token number raises inference latency and GPU memory use.

To investigate the impact of image token count, we conducted an empirical analysis shown in Fig. 7. In page-based retrieval, performance improved with more tokens, leveling off between 384 and 512. Models fine-tuned with the MGF framework showed greater gains above 256 tokens, indicating page-based retrieval benefits from richer visual input. In contrast, for element-based retrieval, dominated by text elements, gains from extra image tokens were relatively smaller.

E. Cross-Language Scenarios

To assess the generalization of MGF-tuned embeddings, we performed experiments in cross-language scenarios, where queries and retrieved items were in different languages. Using GPT-4.1, we translated queries in PCKB between English and Chinese, carefully ensuring domain-specific semantics remained intact. As shown in Fig. 8, model performance generally decreased under cross-language conditions. Nonetheless, MGF-tuned models demonstrated substantial improvements over their nontuned counterparts. An exception was the Jina model, whose performance dropped for image elements in page-based retrieval and for table elements in both settings after MGF-based fine-tuning. This suggests that the MGF's impact may vary with model architectures or modalities, warranting further investigation.

F. Cross-Domain Generalization

Although our data and system are primarily developed for prefabricated construction to support information retrieval and decision-making among multiple stakeholders, it is important to assess whether the trained models maintain their performance when applied to other related industrial domains. Such cross-domain evaluation can demonstrate the robustness and generalizability of models beyond their original application.

In this section, we further evaluate the cross-domain generalization capability of the models using test sets derived from previous studies [8], [20]. Specifically, we assessed model performance across academic papers (sourced from arXiv), government reports, healthcare industry documents, and materials from the energy sector. Table XI presents a comparative analysis between the original models and the MGF-tuned variants. The results substantiate the robustness of the MGF-based fine-tuning approach, demonstrating that the model maintains efficacy even when applied to semantically distinct domains beyond its primary training scope.

G. System Demonstration

Fig. 9 presents the core components of the developed retrieval module in our practical system. Users can query prefabricated construction information from both general and user-uploaded

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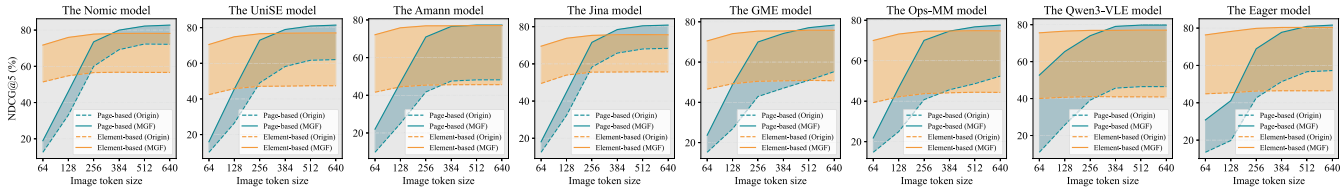


Fig. 7. Comparison of NDCG@5 (%) performance between original and MGF-tuned embedding models across varying image token sizes under page- and element-based retrieval settings.

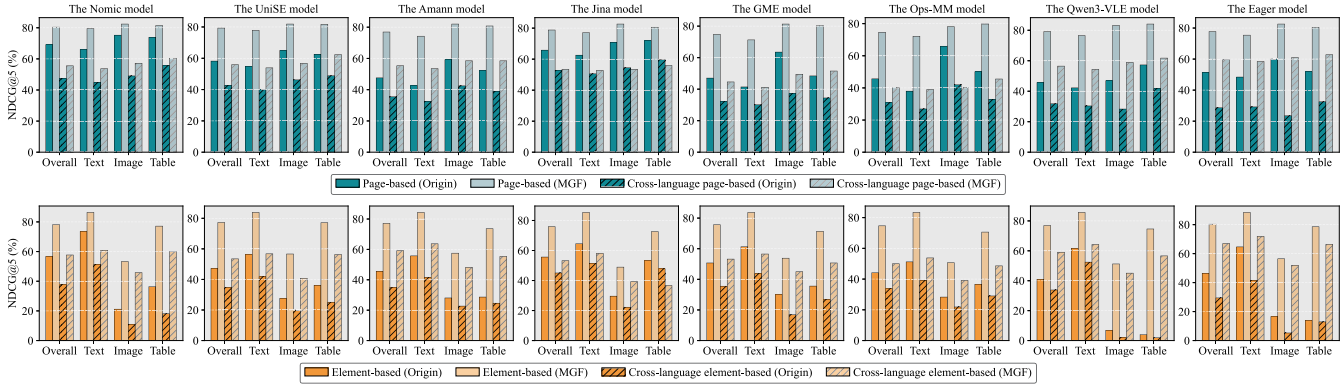


Fig. 8. Comparison of NDCG@5 (%) performance between original and MGF-tuned embedding models under page- and element-based retrieval settings in normal and cross-language scenarios.

TABLE XI
CROSS-DOMAIN GENERALIZATION (NCCG@5) OF ORIGINAL AND MGF-TUNED MODELS ACROSS DIVERSE INDUSTRIAL DOCUMENTS

Model		Academic	Government	Healthcare	Energy
Nomic	Origin	96.55	81.84	82.16	84.25
	MGF	100.0 (↑3.6%)	82.79 (↑1.2%)	83.11 (↑1.2%)	84.59 (↑0.4%)
UniSE	Origin	96.55	80.40	80.40	83.63
	MGF	98.28 (↑1.8%)	81.46 (↑1.3%)	80.67 (↑0.3%)	83.85 (↑0.3%)
Amann	Origin	96.76	78.63	77.28	75.72
	MGF	98.72 (↑2.0%)	83.38 (↑6.0%)	81.78 (↑5.8%)	79.79 (↑5.4%)
Jina	Origin	96.55	80.48	81.29	78.73
	MGF	98.28 (↑1.8%)	82.30 (↑2.3%)	81.32 (↑0.0%)	81.59 (↑3.6%)
GME	Origin	97.89	78.08	78.44	75.74
	MGF	98.28 (↑0.4%)	79.77 (↑2.2%)	80.79 (↑3.0%)	83.49 (↑10.2%)
Ops-MM	Origin	97.00	72.04	71.05	70.19
	MGF	98.28 (↑1.3%)	77.80 (↑8.0%)	78.34 (↑10.3%)	77.14 (↑9.9%)
Qwen3-VLE	Origin	98.28	75.14	77.15	78.82
	MGF	98.28 (↑0.0%)	85.01 (↑13.1%)	83.65 (↑8.4%)	82.03 (↑4.1%)
Eager	Origin	100.0	82.32	78.24	83.54
	MGF	100.0 (↑0.0%)	82.81 (↑0.6%)	81.91 (↑4.7%)	85.90 (↑2.8%)

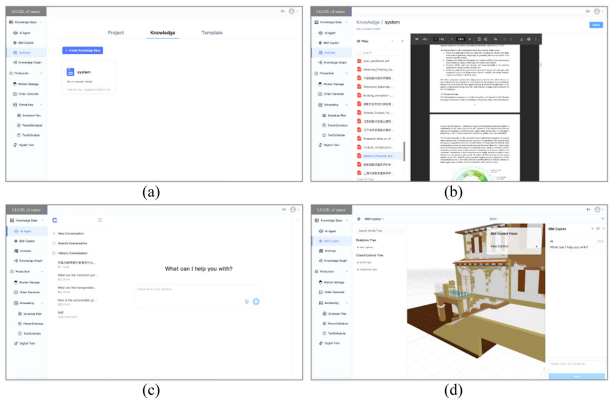


Fig. 9. System demonstration. (a) Interface of knowledge bases. (b) Online document viewer. (c) Interface of AI agent. (d) AI agent in specific application modules.

documents. The PCKB serves as the initial knowledge base to provide foundational information. To enhance knowledge coverage and address user-specific needs, the system also allows users to upload their own documents. The MGF-tuned Nomic model was integrated due to its superior performance. The system operates in two contexts. First, it is accessible via a dedicated functional page where users can query directly. Each query is encoded for retrieving relevant information at both the page and element levels. Then, an LLM synthesizes an answer based on the retrieved knowledge. Second, the assistant is embedded within specific application modules, e.g., model design and production scheduling, providing context-aware support, enabling users to resolve workflow-specific issues in real time.

Documents are processed locally using MinerU, and metadata is stored in a MySQL relational database, while embeddings are indexed in Qdrant,[12] a vector database that supports scalable indexing and efficient similarity search. The system was deployed on a server with an Intel Xeon Platinum 8370 C CPU (2.80 GHz, 32 cores, 64 threads) and an NVIDIA GeForce RTX 4090 GPU with 24 GB memory. Upon uploading, documents are processed concurrently at page and element levels. For instance, a 57-page document with ten images and four tables requires about 22.4 s to segment the document into image-based pages and 38.3 s to parse it into individual elements. The encoding

[12][Online]. Available: https://qdrant.tech/

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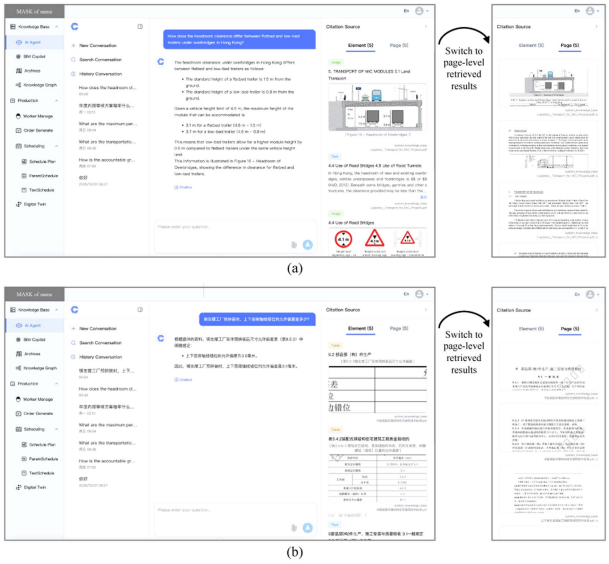


Fig. 10. Cases of retrieval. (a) Case 1. (b) Case 2.

process takes about 0.36 s per page, 0.17 s per image, 0.19 s per table, and 0.09 s per text snippet. During document parsing and encoding inference, the system utilizes about 8 GB of memory. Currently, the processing pipeline runs on a single GPU; overall throughput could be improved by parallelizing across multiple, higher-performance GPUs. Vector-based search over more than 6000 embeddings must take less than 0.01 s. After relevant information is retrieved, the answer generation module uses an external LLM application programming interface (API), whose latency is influenced by network conditions and typically ranges from 0.5 to 10 s. Streaming output and a "thinking" indicator have been implemented to provide a responsive user experience.

Fig. 10 displays two retrieval cases in which users submit queries and receive responses accompanied by results at both the page and element levels.

VI. Discussion

In Section II-D, we identified two challenges hinder deploying multimodal embedding models in prefabricated construction. They manifest at domain and model levels, which we addressed through a codesigned data and modeling strategy.

From the domain perspective, there is an absence of multimodal embedding models tailored to prefabricated construction. Existing models, predominantly developed for general domains, exhibit performance degradation when applied to this specialized domain. Moreover, there is a lack of established datasets in this field. To mitigate these limitations, we assembled prefabricated construction documents and curated a cross-granularity-annotated query–answer dataset. This established pipeline is transferable and can be adapted to other fields.

At the model level, current retrieval models typically treat the page and element levels as independent supervision sources and separate optimization objectives, without explicit alignment, thereby failing to leverage dual-level granularities. To overcome this, we refined our dataset to explicitly include query–answer pairs grounded in aligned page-element pairs, forming the PCKB dataset. Building upon this, we propose the MGF framework, which jointly optimizes models using three contrastive losses to explicitly align multigranularity information. Experiments on five multimodal embedding models demonstrated that MGF-tuned models outperformed both vanilla and granularity-specific fine-tuned models.

Nonetheless, limitations exist. First, our system did not implement a reranking module (e.g., page-based retrieval for recall and element-based retrieval for precision), as empirical observations showed that both granularities yield comparable relevance metrics and appear to offer complementary semantic insights to end users. Further theoretical analysis is needed to explore these interactions in greater depth. Second, the models exhibited suboptimal performance under certain conditions, e.g., cross-language scenarios. Addressing these limitations may require more embedding strategies, e.g., multivector or late fusion approaches, which go beyond the standard dense embedding scope in this study; such enhancements are therefore reserved for future research. Third, our evaluation focused on the prefabricated construction domain, which involves multiple stakeholders from diverse disciplines, and broader validation through additional user studies is warranted. We will continuously expand the collection of documents by updating our code repository accordingly to enhance coverage. Furthermore, further research is needed to generalize and validate its effectiveness across additional domains, thereby benefiting a broader spectrum of industry applications.

VII. Conclusion

This work focuses on advancing domain-specific dense embedding models that support both page- and element-based multimodal retrieval for prefabricated construction. To support this effort, PCKB, a domain-specific dataset comprising query-answer pairs at both the page and element levels, was curated. The established data curation pipeline serves as a reference for future research in related areas. Based on this, the MGF framework is proposed, which explicitly integrates multigranularity information to improve multimodal embedding models. Empirical evaluations validate the effectiveness and potential of MGF. To bridge the gap between research and practice, the proposed technique was integrated into a functional system designed for real-world deployment in prefabricated construction workflows. This work not only offers a generalizable framework for multimodal retrieval systems in other domains but also provides valuable insights into the benefits of fusing page- and element-level information for more meaningful representations. Limitations include the lack of reranking, suboptimal performance in certain scenarios, and limited user-centric evaluation within and beyond the prefabricated construction domain. Future directions include deeper theoretical analysis, exploration of more embedding strategies, user-centered evaluations, and extension to broader domains to fully realize the potential and impact.

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